Abdullah Al Mamun

Software Engineer

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Github Linkedin Portfolio



PROFESSIONAL SUMMARY

Entry-level Software Engineer and Computer Science and Engineering (CSE) graduate with hands-on experience in **full-stack web development** and **machine learning**. Builds and deploys web applications and RESTful APIs using JavaScript, Next.js, React, Node.js, Express.js, FastAPI, PostgreSQL and Prisma ORM, with JWT authentication and role-based access control. Fine-tuned Whisper and BanglaT5 models with Python, PyTorch and Hugging Face Transformers for Bengali speech recognition and text summarization (NLP), reducing word error rate from **55.0% to 17.3%**. Skilled in Git, GitHub and cloud deployment (Vercel, Render); coordinated technical events for 100+ participants. Seeking a **Junior Software Engineer** or **Full-Stack Developer** role.

PROJECTS

- **Bengali Audio Transcription and Summarization with a Newly Developed Dataset** Github
  Thesis Project | Python, PyTorch, Hugging Face Transformers, Whisper-small, BanglaT5, FastAPI, FFmpeg, Kaggle
  · Dataset: developed a new Bengali speech dataset of 1,160 audio clips with verbatim transcripts and reference summaries (928 train / 116 validation / 116 test).
  · Speech recognition: fine-tuned Whisper-small, reducing word error rate (WER) from 55.0% to 17.3% and character error rate (CER) from 22.6% to 6.0% on the test set.
  · Summarization: fine-tuned BanglaT5 for abstractive summarization, improving ROUGE-L from 14.5 to 25.3 over the earlier mT5-small model.
  · Features: FastAPI web app that converts uploaded Bengali audio into a transcript and a concise summary, with pause-based audio chunking.
- **Scholario – Assignment & Submission Management System** Live Demo | Github
  Full Stack | Next.js, Node.js, Express, PostgreSQL, Prisma ORM, JWT, bcrypt, Axios, React Context API
  · Built a full-stack assignment and submission management platform supporting student, teacher and admin roles with JWT-based authentication and bcrypt password hashing.
  · Developed a RESTful backend with Express and Prisma ORM on PostgreSQL (Neon), covering assignment CRUD, submission lifecycle, grading with feedback and role-based access control.
  · Added email verification and password reset via the Gmail API (OAuth2); deployed the frontend on Vercel and the backend on Render.
- **StudyVault – An Online Education Platform**
  Minor SDP Project | PHP, JavaScript, CSS, MySQL, GitHub
  · Built a web-based educational platform enabling students to access and manage online courses.
  · Implemented RESTful routing with a PHP backend and MySQL database; delivered complete user authentication, course listing and content management features.

SKILLS

- **Programming Languages**
  C, C++, Python, JavaScript, PHP, SQL, HTML/CSS
- **Frameworks & Libraries**
  Next.js, React, Node.js, Express.js, FastAPI, Prisma ORM, Hugging Face Transformers, PyTorch, Django (familiar), OpenCV

- **AI / ML**
  Machine Learning, Natural Language Processing (NLP), Speech Recognition (Whisper), Text Summarization (BanglaT5, mT5), Model Fine-Tuning, Evaluation (WER, CER, ROUGE)
- **Databases**
  PostgreSQL, MySQL, Firebase
- **APIs & Integration**
  REST APIs, HTTP protocols, JWT Authentication, Gmail API (OAuth2), Google Speech API, Webhooks (basic)
- **Developer Tools & Platforms**
  Git, GitHub, VS Code, Figma, Google Colab, Kaggle, Vercel, Render
- **Languages**
  Bengali (Native), English (Professional – written & verbal)

EDUCATION

- **BSc in Computer Science and Engineering** (2022 – 2026)
  Bangladesh University of Business and Technology (BUBT)
- **Higher Secondary Certificate** (2020)
  Govt. Bangla College
  GPA: 5.00 / 5.00
- **Secondary School Certificate** (2018)
  Mirpur Bangla High School
  GPA: 4.83 / 5.00

EXTRACURRICULAR ACTIVITIES

- **Video Editor** – BUBT IT Club (2025 – 2026)
- **Event Coordinator** – BUBT Intra University Programming Contest (2022 – 2024), 3 contests, 100+ participants
- **Event Coordinator** – INNOVATEX Hackathon, BUBT IT Club (2024)
- **Volunteer** – ICPC Asia Dhaka Regional Contest (2021, 2023)
- **2nd Runner Up** – BUBT Leader of The Year, Video Editing Segment (2024)

HOBBIES AND INTEREST

- Travel Filmmaking (completed 32 districts of Bangladesh)
- Videography, Cinematography and Creative Visual Storytelling (Instagram: @foyez_gone_far)